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A Chaotic Gradient-based Optimization with Support Vector Machine for Chinese Folk Music Classification

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Abstract—**Nowadays, Chinese folk music classification has a huge attraction on the researches end where traditional Chinese folk music classification has difficulty with feature selection, and current Machine Learning (ML) algorithms have limitations such as poor classification accuracy. Conventional music classification algorithms struggle to accurately classify Chinese folk music due to its sensitive rhythms and tone changes. The goal is to develop algorithms that can accurately recognize and classify diverse folk tunes in the Chinese music collection. So, Chaotic Gradient-Based Optimizer (CGBO) is introduced that merges chaotic maps with GBO searching rounds to find the most discriminating features and finally for folk music categorization, the Support Vector Machine (SVM) algorithm is used. In this research to evaluate the performance of the proposed CGBO-SVM, input data is created by self as set of ethnic music data, this data collection includes 10 different genres. Every genre includes minimum count of 100 samples, and every sample lasting 30 secs with sampling rate of 22050 Hz, while the One Note detector technique is utilized to effectively extract features. The experimental results reveal that the suggested CGBO-SVM achieve 99.96% of classification accuracy and performs better than existing approaches such as K-Nearest Neighbor (KNN) and Bi-Directional Long Short Term Memory (Bi-LSTM).**

Keywords—chinese folk music classification, machine learning, chaotic gradient-based optimizer, support vector machine, one note detector.

I. INTRODUCTION

As similar to art, Music communicates people's ideas, emotions, and lifestyles, as well as promote emotional and spiritual well-being [1]. The emergence of Music has suddenly rise and become popular in people's circumstances. As science and technology advanced, an increasing number of individuals preferred to listen music over the Internet. As a result, determining which music people want to listen to from a large collection of music has become more crucial, and music identification and categorization have received increased attention. Particularly, the examples of Chinese folk songs are Instrumental music, songs, and opera [2] which comprises a diverse set of essential forms and approaches based on Chinese culture. Despite its broad reach and historical importance, academics encounter a number of challenges when classifying Chinese folk music [3]. The intricate structure of Chinese folk music is a difficulty for traditional algorithms, thus resulting in erroneous classification outputs. Folk music relies heavily on musical instruments where it may be played individually or in an ensemble, and various combinations of musical instruments create diverse kinds of instrumental music. The purpose of this study is to bridge the gap between old forms of Chinese folk music and modern technology by classifying Chinese folk music. Using Machine Learning (ML) technologies [4-5], researchers aimed to preserve the cultural heritage of Chinese folk music by developing powerful and effective categorization systems. ML is a subfield of AI which focuses on creating models that learn from data and utilize that knowledge for classification. SVM is a popular ML technique for regression and classification issues. However, without adequate feature selection, SVMs have trouble in classifying both high-dimensional and nonlinear data. So, the CGBO combines chaotic maps with GBO search rounds to find the most discriminating features. This study highlights the effectiveness are utilized to protect cultural heritage, and it represents a significant development in Chinese folk music categorization systems. Kostrzewa et al [6] suggested that music genre recognition be done by using the capabilities of large ensembles of neural network classifiers. Researchers have made significant contributions to the progress of automated musical genre identification by creating uncommonly utilised forms of ensembles. By incorporating large ensembles, researchers get another approach for enhancing the classification outcome without requiring significant design or programming work. A limitation might be higher calculation time as well as demand for computer resources.

Wijaya et al. [7] proposed a customized BiLSTM model with Mel Frequency Cepstral Coefficients (MFCC) feature extraction. BiLSTM was a neural network design that enables the model to acquire patterns in both directions. To get normalized input, preparatory techniques such as silent portion removal and stretching are undertaken. The model was quite excellent at categorizing positive occurrences of the "pop" class; but, the model tends to overlook certain positive examples that should be categorized; this might be owing to the relatively short sample size, which was just six records. Zhang [8] proposed a unique technique that uses music extraction as well as KNN to solve the issue of poor accuracy in existing methods. As a consequence, a music categorization module based on a KNN's one-dimensional convolution was suggested , which paired with a two-way KNN network. The output was weighted differently to properly reflect the music style features. The tonal feature was employed to achieve performance second only to the suggested approach, however the melody element has a higher influence on performance. Ceylan [9] presented a Convolutional Neural Network (CNN) to categorize music genres, based on the prior successful findings. Instead of utilising visual characteristics or representations, music segments in the dataset were represented using MFCC. We obtain MFCCs by preprocessing the dataset's music pieces, then train a CNN model using the MFCCs and assess the model's success using the testing data. A definite decision cannot be formed since certain instances in deep learning remain unexplained. Elbir

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[10] proposed DeepMUSIC, a DL system for multiple signal classification. Each CNN acquires the MUSIC spectra of its own angular subregion. The algorithm created a non-linear link between sensor data and the angular spectrum. While DeepMUSIC includes numerous networks in comparison to Multi-Layer Perceptron (MLP), it gives less computing time because DeepMUSIC has several convolutional layers rather than fully linked layers, which require greater complexity.

The primary contributions of the research are as follows:

- The CGBO combines chaotic maps with GBO iterations to select discriminating characteristics and improve accuracy in Chinese folk music classification.
- The self-created ethnic music data collection consists of ten genres, each with 100 audio samples lasting 30 seconds where the sampling rate is 22050 Hz, and from the data features are extracted using the Note onset detector technique.
- Finally, for effective Chinese folk music categorization, the SVM is used which classifies optimum features.

The remaining of the paper is: The proposed methodology is briefly explained in section 2 whereas the experimental results are detailed in section 3. At last, the conclusion of the overall research is presented in section 4.

II. Proposed Methodology

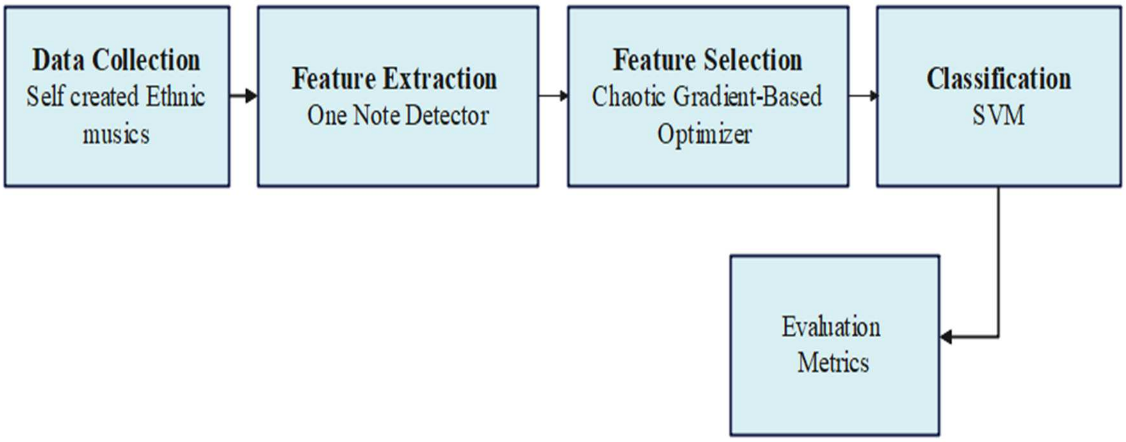

Fig. 1. Block diagram of the proposed methodology

In this research, CGBO that combines chaotic maps with searching iterations of the GBO for selecting most discriminating features to achieve high performance in Chinese folk music classification. As shown in Figure 1, the overall procedure followed in this research is described in the following sections clearly.

A. Data Collection

In this research to evaluate the performance of the proposed CGBO-SVM, input data is created by self as ethnic music data this data collection includes 10 different genres. Every group includes minimum count of 100 samples, where an every sample lasting 30 secs. Whereas 100 audio samples sampling rate is 22050 Hz [11]. The collected input samples are given to one note extraction technique for feature extraction.

B. Feature Extraction

Note onset detector extraction is the process of finding the exact times in audio data that musical notes originate. This method monitors variations in spectral content over time. Notation onsets are associated with fast signal shifts (Melspectrum features) that are captured by note onset extraction. Note onset times are accurately determined by using time-domain properties including short-time energy and amplitude envelope where the extracted features are given as inputs to CGBO for feature selection.

C. Feature Selection (FS)

From the extracted features, FS allows picking a feature subclass having necessary information and redundancy free thus enhances categorization performance substantially. Through integrating greatly associated characteristics/ features with additional characteristics, ML techniques improve classification accuracy. Eventhough if two characteristics have a great connection, only one is employed to correctly describe the data.

1) Gradient-Based Optimizer (GBO): GBO [12] is a strategy for addressing hard optimisation issues that combines population-based and gradient approaches. The GBO method uses Newton's technique to control the direction of the search agent while it explores the problem space as shown in Eq (1). The Gradient Search Rule (GSR) along the Locale Escaping Operator (LEO) are two key factors of GBO method.

$$X_n = X_{min} + rand(0,1) \times (X_{max} - X_{min}) \tag{1}$$

2) GSR Phase : GBO technique begins with group of initial solutions it updates each agent's location in a gradient-based location. The crucial factor is employed to establish a balance between investigating substantial search space regions as well as exploitation to get near-optimal solutions.

$$GSR = randn \times \rho_1 \times \frac{2\Delta x \times x_n}{(x_{worst} - x_{best} + \varepsilon)} \tag{2}$$

GBO does a randomized exploration, which involves locating local optima utilizing random behaviour. The above equation, Eq. (2), demonstrates how to calculate the random offset that represents the variance among optimal answer and random chosen answer. In exploration stage, additional random number is in eqs (3-5):

$$\rho_1 = 2 \times rand \times \alpha - \alpha \tag{3}$$

$$\alpha = \left|\beta \times sin\left(\frac{3\pi}{2} + sin\left(\beta \times \frac{3\pi}{2}\right)\right)\right| \tag{4}$$

$$\beta = \beta_{min} + (\beta_{max} - \beta_{min}) \times \left(1 - \left(\frac{m}{M}\right)^3\right)^2 \tag{5}$$

3) LEO Phase: LEO is introduced to improve the suggested GBO's efficacy in solving complex issues where the LEO can successfully update the solution's location. It helps GBO break out of local optimum spots as well as accelerate convergence. As shown in Eq (6), LEO gives a solution with greater efficacy by merging many solutions,

$$\begin{aligned} &If\, rand < pr \\ &X_{LEO}^m = \begin{cases} X_n^{m+1} + f_1(u_1 x_{best} - u_2 x_k^m) \\ \quad + f_2\rho_1\left(u_3(X2_n^m - X1_n^m)\right) \\ + u_2(x_{r1}^m - x_{r2}^m)/2,\ if\, rand < 0.5 \\ X_n^{m+1} + f_1(u_1 x_{best} - u_2 x_k^m) \\ \quad + f_2\rho_1\left(u_3(X2_n^m - X1_n^m)\right) \\ + u_2(x_{r1}^m - x_{r2}^m)/2,\ otherwise \end{cases} \\ &End \end{aligned} \tag{6}$$

Specifically, the CGBO's stages are carried out as: To preserve a balance among exploitation & exploration, CMs are employed which helps to prevent local optima as well as early convergence, as well as; SVM is subsequently utilized for classification. The suggested technique's solutions must be examined at every iteration to verify that they function as predicted. The fitness function ($fobj$) of CGBO is described in Eq (7):

$$Fitness\ function\ (fobj) = \alpha + \beta\frac{|R|}{|C|} - G \tag{7}$$

SVM is a quadratic method employed in this research to accurately classify Chinese folk music. The SVM classifier's purpose is to generate "good" separation hyperplanes between classes in a high-dimensional feature space in a computationally efficient way. The input vectors are turned into a high-dimensional feature space by using nonlinear transformation as well as linear separation. To produce a nonlinear SVM classifier, the inner product is replaced by a kernel function, as indicated in Eq (8),

$$f(x) = sgn\left(\sum \alpha_i y_j K(x_i, z) + b\right) \tag{8}$$

A SVM classifier's fundamental form is as in Eq (9):

$$Y(z) = w^T \Phi(z) + b \tag{9}$$

The feature mapping function result is the feature space $\Phi(z)$.

From (7) and (8), the test data z of SVM is in eq (10),

$$Y(z) = \sum_{sv=1}^{N} \alpha_{sv} y_{sv} (\Phi^T(\chi_{sv})\Phi(z) + b = \sum_{sv=1}^{N} \sum_{sv} y_{sv} K(\chi_{sv}, z) + b \tag{10}$$

From this, it clearly demonstrates that the SVM obtain higher classification accuracy in Chinese folk music classification.

III. Experimental results

In this section, using the MATLAB 2020 software, Intel i7, 8GB RAM, 32 bit processor, the performance of the proposed CGBO-SVM was tested. The purpose of the simulation experiments is to confirm that the CGBO-SVM approach works effectively. To evaluate the proposed CGBO-SVM such as classification accuracy and error rate where the mathematical equations is given in Equations (11 and 12).

$$Classification\ Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \tag{11}$$

$$Error\ rate = \frac{FP+FN}{P+N} \tag{12}$$

Where, TP-True Positive, FP – False Positive, TN-True Negative, FN- False Negative.

TABLE I. Performance analysis of the proposed CGBO with traditional approaches

Methods	Classification Accuracy (%)	Error rate (%)
GBO	93.54	13.07
BOA	95.55	8.16
WOA	96.22	4.34
CGBO	99.96	1.17

The performance of the CGBO is analyzed with the traditional approaches such as Bat Optimization (BOA), Whale Optimization Algorithm (WOA) to determine its efficiency whereas the obtain results are tabulated in table 1. The analysis is conducted using the considered performance measures such as recognition accuracy and Error rate.

From the above table, it clearly demonstrates that the proposed method obtains a higher recognition and data loss of 99.96% and 1.17%. It clearly demonstrates that the CGBO outperforms the traditional approaches. This superior performance is achieved through the utilization of CGBO for feature selection process. Thus, the CGBO-SVM model shows high level of effectiveness in Chinese folk music classification.

The comparison analysis between proposed CGBO with conventional models is conducted. The enhanced performance of the CGBO-SVM is determined by comparing it with the conventional approaches such as KNN [8], Bi-LSTM [7] to determine its efficiency and the obtained results are tabulated in table 2. The comparison is conducted using classification accuracy.

TABLE II. Comparison evaluation of the proposed with existing approaches

Models	Classification Accuracy (%)
KNN [8]	95.4
Bi-LSTM [7]	94.60
CGBO-SVM	99.96

From Table 2, it clearly demonstrates that the CGBO-SVM obtains a higher accuracy of 99.96% and outperforms the existing works. Whereas the KNN [8] and Bi-LSTM [7] obtains an 95.4% and 94.60% respectively. This superior performance is achieved through the utilization of CGBO for feature selection process. Thus, the CGBO-SVM model shows high level of effectiveness in Chinese folk music classification

A. Discussion

In this research, a CGBO method is proposed for the SVM's hyperparameter optimization to achieve superior classification accuracy. A set of self-created music data collection is used to evaluate the CGBO-SVM and One Note detector approach is used to efficiently extract features from the input data whereas the advantages of the proposed approach are discussed. The following results demonstrate the efficiency of the CGBO-SVM in successfully forecasting the data patterns. The CGBO-SVM obtains a higher accuracy of 99.96% indicating better overall performance contrasted to traditional approaches whereas the KNN, Bi-LSTM obtains 95.4% and 94.60% respectively. This superior performance is achieved through the utilization of CGBO for feature selection process. Thus, the CGBO-SVM model shows higher performance in Chinese folk music classification

IV. CONCLUSION

The main objective of this research is to create model for precise classification of Chinese folk music. So, a CGBO is proposed for feature selection to select optimal features from the extracted feature, thus helps SVM to achieve superior classification accuracy of Chinese folk music. A self-created music collection is used to evaluate the CGBO-SVM and One Note detector approach is used to efficiently extract features from the input data. The CGBO-SVM obtains higher classification accuracy of 99.96%. Thus, the CGBO-SVM model shows high level of effectiveness in Chinese folk music classification. The future work will be focused on utilizing hybrid optimization techniques to improve the recognition accuracy of the DL models.

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